

LOANZO PROTOCOL

DECENTRALIZED PEER-TO-PEER LENDING WITH BIOMETRIC LEGAL ESCROW

HACKATHON & STARTUP DOSSIER

v2.5 Full-Proof Master Edition • 2026

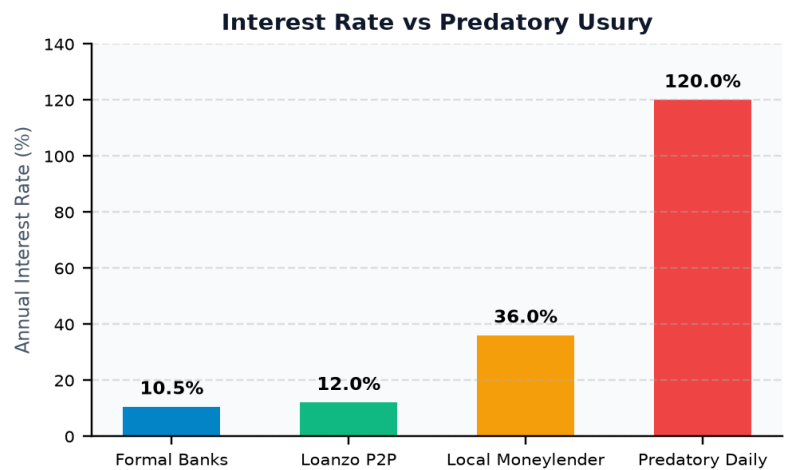
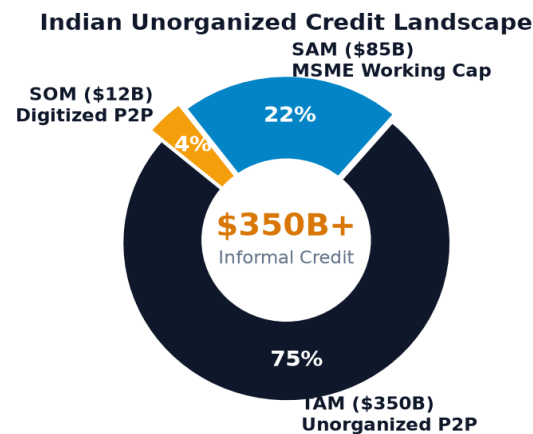


MEET THE PRIMARY BORROWER: ROHAN SHARMA

Profile: Electronics retailer in Pune needing INR 5,00,000 festival stock inventory.

- **The Dilemma:** Local moneylenders demand 24%–36% usurious compounding interest. Commercial banks take 3–4 weeks.
- **The Solution:** Rohan registers on **Loanzo** using 30-second DigiLocker e-KYC and posts an asset-backed request at 8.2%.
- **Co-Borrower:** Rohan adds sister **Priya Sharma (@priya)** for joint liability authorization.
- **Result:** Connects with Community Lender **Aisha Khan**, saving INR 45,000 in interest!

MARKET OPPORTUNITY & PROBLEM-SOLUTION BENCHMARK



THE INFORMAL CREDIT CRISIS (\$350B TAM)

- **Zero Legal Enforceability:** Informal handshakes lack promissory note validity under the NI Act 1881.
- **Predatory Usury:** Compounding daily rates reach 36% to 120% per annum.
- **Muscle Intimidation:** Disputes trigger harassment without legal civil recourse.
- **Zero Credit Build-up:** Timely repayments build zero official CIBIL footprint.

THE LOANZO PROTOCOL REVOLUTION

- ✓ **Binding Promissory Notes:** 4-page automated contracts under Section 4, NI Act 1881.
- ✓ **Doorstep Asset Valuation:** Calibrated gold touchstone assaying by certified agents.
- ✓ **Direct Wholesaler Tranche:** Capital paid directly to distributors to stop diversion.
- ✓ **Instant Legal Clearance NOC:** Formal debt satisfaction letters upon zero balance.

DOORSTEP VERIFICATION & COLLATERAL VALUATION MESH



FIELD OPERATIONS OFFICER: VIKRAM RAO (@AGENT)
Equipment: Calibrated electronic jeweler's scale, touchstone acid assaying kit, Loanzo Field Agent App.
• **Doorstep Dispatch:** Rohan pledges 50g family gold jewellery. Loanzo dispatches Agent Vikram within 30 mins.
• **Purity & Weight Verification:** In front of Rohan & Priya, Vikram verifies 50.1g solid 22K gold.
• **Live IBJA Price Matching:** The Agent App fetches real-time gold market rates, establishing INR 3.5 Lakh asset backing.
• **Tamper-Evident Security Seal:** Vikram packs the ornaments in security pouch with barcode #G408459.

Loanzo 3-Tier Multi-Role Ecosystem Architecture



FIELD AGENT APP & GEOPROCESSING TECHNICAL SPECIFICATION

FEATURE	IMPLEMENTATION ARCHITECTURE	RISK MITIGATION
GPS Geofencing	FusedLocationProviderClient (< 50m radius)	Prevents ghost / fabricated inspection reports
Tamper Pouch Barcode	ZXing Barcode scanning + SHA-256 hash	Eliminates collateral substitution or theft
Live IBJA Rate Sync	Automated Bullion API via WorkManager	Protects LTV ratios against gold price volatility
Offline Visit Logging	Room DB v12 SQLite transactional queue	Enables visits in rural Tier 2/3 zero-signal zones

COLLATERAL VAULT ESCROW & LEGAL MULTI-PARTY ESIGN



CENTRALIZED COLLATERAL VAULT & ESCROW SYSTEM
Master Admin Module: Overseen directly by App Owner Node (@satyam0810).
• **Physical Locker Allocation:** Pouch #G408459 is deposited in high-security Regional Locker BLR-VAULT-208.
• **Dual-Custodian Release Protocol:** Collateral cannot be moved without dual biometric sign-off from Master Admin and Valuer.
• **Direct Wholesaler Tranches:** INR 2,00,000 is transferred directly to the verified distributor UPI (distributor@bank) to eliminate borrower cash diversion.
• **Automated Lien Custody Tracking:** Full audit trail visible to both Borrower and Lender in app.

Loanzo Complete 5-Stage Lending Lifecycle (From Comic Book Storyboard)



1. Proposal & Vouch 2. Gold Valuation 3. Multi eSign 4. Direct Tranche 5. Golden NOC

Rohan posts INR 5L request 18 market vouches (4.9) Agent Vikram tests 22K Seals pouch #G408459 Rohan, Priya & Aisha sign Disbursed via UPI to distributor Final EMI paid (INR 0 dues) NI Act Promissory Note Rule Engine stops diversion Gold returned + Legal NOC

LEGAL BINDING FRAMEWORK (INDIAN NEGOTIABLE INSTRUMENTS ACT 1881)

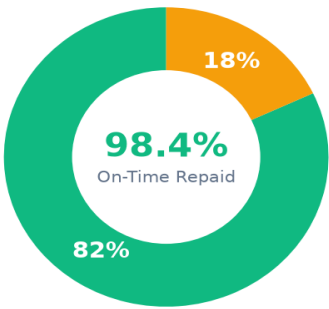
STATUTORY PROVISION	LOANZO IMPLEMENTATION	LEGAL CONSEQUENCE
Section 4, NI Act 1881	Unconditional Promissory Note executed by Rohan & Priya	Direct cause of action for summary civil debt recovery
Section 65B, Evidence Act	SHA-256 digital stamp + device IP + timestamp cert	Admissible electronic proof without secondary witnesses
Section 138, NI Act 1881	NACH / UPI mandate authorization for EMI debits	Criminal liability for intentional dishonour of mandate
RBI Digital Directions	Direct P2P UPI rail transfer without pool account	Zero platform insolvency risk or capital misappropriation

RECOVERY PERFORMANCE, ZERO BALANCE & OFFICIAL NOC

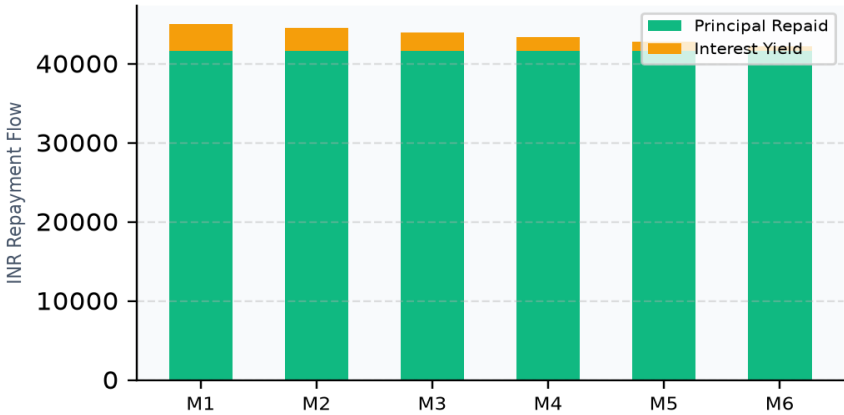


- FULL DEBT SATISFACTION & GOLDEN NOC ISSUANCE
- Lender: Aisha Khan • Borrower: Rohan Sharma • Co-Borrower: Priya Sharma
- **Punctual EMI Track:** 12 EMIs paid on time via **SwipeToConfirmButton**.
 - **Credit Health Surge:** Both Rohan's and Priya's scores rise to **820 (Prime Tier)**.
 - **Collateral Safely Returned:** Sealed gold pouch #G408459 handed back to Rohan in pristine condition.
 - **Golden NOC Issued:** Loanzo generates an official, tamper-proof **No Objection Certificate (NOC)** and debt clearance satisfaction letter (PDF).

Capital Recovery Ratio



Rohan's 6-Month Amortization Trajectory



COMPLETE 20-PANEL GRAPHIC NOVEL MAPPING & RECOVERY ENGINE

PANEL #	STORY MILESTONE	LOANZO ARCHITECTURAL COMPONENT	SYSTEM STATE
Panels 1–4	Diwali Crunch & DigiLocker KYC	Aadhaar/PAN OCR + DigiLocker SDK	DRAFT → VERIFIED
Panels 5–8	Marketplace Bids & Gold Pledge	Bidding Engine + Valuer Dispatch Queue	BIDDING → ASSAY_PENDING
Panels 9–12	Assaying & Multi-Party eSign	Agent Scale Sync + Canvas E-Sign	APPRAISED → ACTIVE
Panels 13–16	Telegram Bot & RBI 2% Penalty	@Loanzo_bot + PenaltyEngine (Grace Days)	ACTIVE / OVERDUE_MONITORED
Panels 17–20	Direct Tranche & Golden NOC	Direct UPI Rail + AgreementGenerator NOC	REPAID → ZERO_DUES_CLOSED

MONETIZATION MODEL, UNIT ECONOMICS & EXPANSION ROADMAP

REVENUE STREAM	FEE STRUCTURE	GROSS MARGIN	10,000 LOANS RUN-RATE
1. Origination Fee	1.5% – 2.5% of disbursed capital	92%	INR 1,25,00,000 (\$150,000)
2. Field Valuer Fee	Flat INR 350 per doorstep visit	35% platform cut (65% Agent)	INR 12,25,000 (\$15,000)
3. Collateral Vault Custody	0.5% Annual LTV on physical gold	80%	INR 25,00,000 (\$30,000)
4. Dispute Mediation Tribunal	Flat INR 500 per virtual hearing	70% platform cut	INR 5,00,000 (\$6,000)

12-MONTH STRATEGIC EXPANSION ROADMAP

Q1 2026: CORE PROTOCOL Status: Completed ✓ • Jetpack Compose M3 Golden UI • Native Canvas Visual Charts • Biometric E-Sign & Selfie • Room DB Offline First Sync • Cross-role Demo Seeder	Q2 2026: AGENT MESH Status: In Progress ■ • GPS geofenced dispatch • Physical barcode scanning • Cash collection receipts • Telegram Bot Bridge API • SMS Token Fallbacks	Q3 2026: REGIONAL EXP Status: Planned ➔ • Launch in UP, Bihar & MH • DigiLocker direct link • Physical locker escrow hub • Regional languages voice AI • Virtual dispute tribunal	Q4 2026: INSTITUTIONAL Status: Roadmap ■ • P2P NBFC syndication • CIBIL / Experian reporting • Securitized P2P notes • Cross-border remittances • Automated court filings
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TECHNICAL ARCHITECTURE & ZERO-BLOAT SPECIFICATION

LAYER	TECHNOLOGY SELECTION	JUSTIFICATION & HARDENING
UI & Presentation	Kotlin 2.0 • Jetpack Compose M3	Unidirectional Data Flow (UDF), 60 FPS fluid rendering, Golden Coin theme
Offline-First Store	Room Database v12 • SQLite	Ensures zero data loss during Tier 2/3 rural connectivity blackouts
Background Sync	Android WorkManager SyncWorker	Queues mutations in SyncQueueEntity; flushes to Cloud when online
Hardware Security	Android BiometricPrompt & KeyStore	Hardware-backed biometric encryption for legal promissory notes
Notification Bridge	Telegram Bot Webhook Engine	Instant multi-channel OTP alerts, dynamic UPI QR codes & EMI countdowns

EXECUTIVE LEADERSHIP & PROTOCOL AUTHOR

Satyam Kumar (@satyam0810) — Founder & Chief Protocol Architect
Engineered for the National FinTech Innovation Hackathon & Seed Stage Venture Acceleration.
Repository: Loanzo Protocol • Master Admin & Owner Node Active.

★ OFFICIAL AUDIT SEAL ★
LOANZO SMART PROTOCOL
✓ NI ACT 1881 ENFORCEABLE
✓ SHA-256 VERIFIED LEDGER